



Federal Urdu University
of Arts, Science and Technology

Artificial Intelligence

Assignment

Intelligent Agent: Smart Finder Agent

Tayyaba Irshad

BSCS - VI | Section: B

Course Instructor: Dr. Uzma Afzal

Intelligent Agent: Match Found AI Agent

1. Identify the Problem

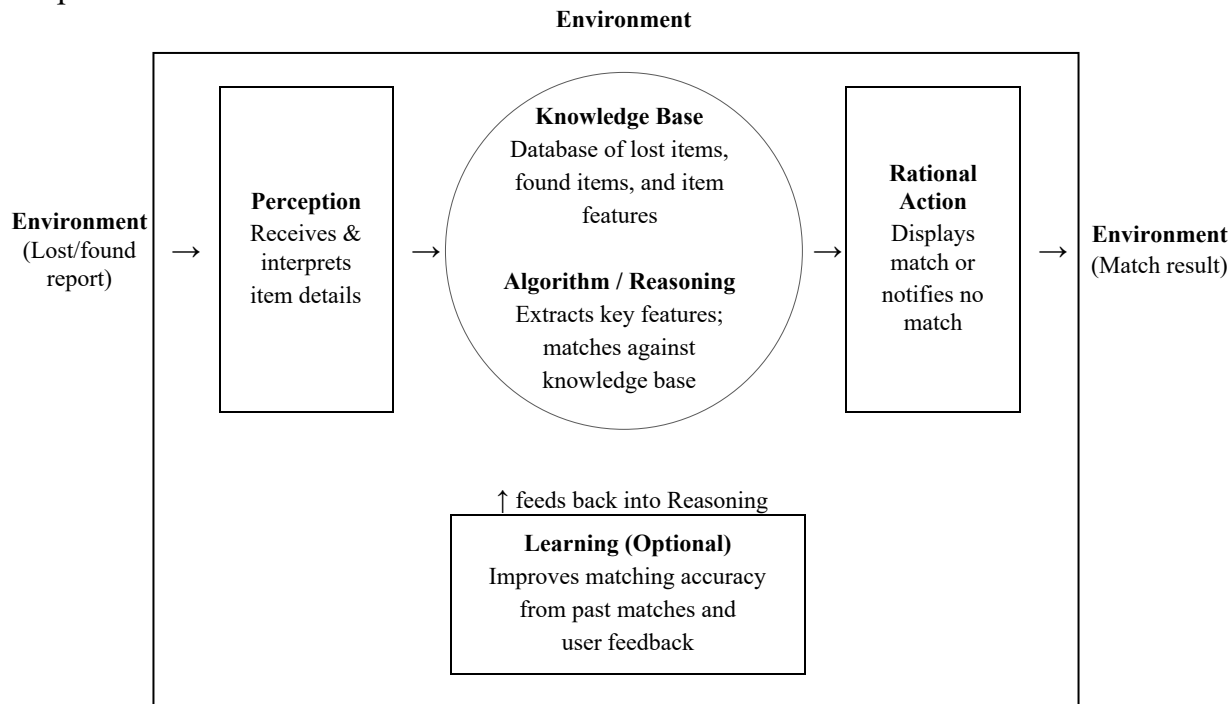
People often lose personal belongings in shared spaces such as campuses or offices, and there is no efficient way to match these lost items with items that have been found. Manually comparing lost and found reports is slow and unreliable. An automated system is needed to determine whether a reported lost item matches a found item.

2. Formulate the Problem

Textual

The system compares the details of lost and found items such as type, color, brand, location, and description to determine whether they match. It then outputs a decision: Match or No Match.

Graphical



3. Evaluate in the Context of PEAS

- **Performance measure:** Accuracy of matches, few errors, speed, and user satisfaction.
- **Environment:** Users reporting items, the item records, and the locations where items are lost or found.
- **Actuators:** Showing match results to the user and updating the records.

- **Sensors:** The forms where users enter item details, and photos if provided.

4. Identify Environmental Properties

- **Partially observable:** The agent only knows what users report; it has no way to verify ground truth beyond the given descriptions.
- **Stochastic:** Outcomes depend on variable, sometimes ambiguous human-entered descriptions, so a match decision is not guaranteed to be certain.
- **Episodic:** Each lost/found comparison is handled largely on its own; the current decision does not depend on a planned sequence of future actions.
- **Dynamic:** New items are reported as lost or found continuously while the system is running.
- **Discrete:** Item types, colors, brands, and locations fall into a finite, discrete set of categories.
- **Single-agent:** The system operates independently; it is not competing with or reacting to other agents.